

# Linux Advanced — Chapter 14

*Tmux: Terminal Multiplexer | Scott Pattison*

**Goal:** Keep terminal sessions alive and organise your workspace with multiple panes and windows.

## 11.1 Why Tmux?

### The Problem

Imagine running a long process over SSH. Your connection drops. Without tmux, the process dies and you have to start over. Or you need to watch logs in one terminal, edit files in another, and run commands in a third — without tmux that means three separate SSH connections.

### The Solution

Tmux solves both problems:

- **Sessions persist** — disconnect and reconnect without losing work
- **Multiple panes** — split your terminal into sections
- **Multiple windows** — like tabs in a browser

## 11.2 Getting Started

### Install Tmux

```
sudo apt install tmux
```

### Start Tmux

```
tmux
```

You're now inside a tmux session. Notice the green bar at the bottom — that's the status line.

### The Prefix Key

All tmux commands start with the prefix: **Ctrl+b**. Press Ctrl+b, release both keys, then press the command key. For example, to detach: **Ctrl+b then d**.

## 11.3 Sessions

| Action                     | Command          |
|----------------------------|------------------|
| Detach from session        | Ctrl+b d         |
| List sessions              | tmux ls          |
| Attach to most recent      | tmux attach      |
| Attach to specific session | tmux attach -t 0 |
| Create named session       | tmux new -s work |

|                |                                        |
|----------------|----------------------------------------|
| Attach by name | <code>tmux attach -t work</code>       |
| Kill a session | <code>tmux kill-session -t work</code> |

**Tip:** Named sessions are clearer than remembering session numbers. Always name your sessions:  
`tmux new -s project`

## 11.4 Windows

Windows are like tabs — each window is a full terminal. The status bar shows your windows: `0: bash`  
`1: vim` `2: logs`

| Action            | Keys                                           |
|-------------------|------------------------------------------------|
| Create new window | <code>Ctrl+b c</code>                          |
| Next window       | <code>Ctrl+b n</code>                          |
| Previous window   | <code>Ctrl+b p</code>                          |
| Window by number  | <code>Ctrl+b 0-9</code>                        |
| Rename window     | <code>Ctrl+b ,</code>                          |
| Close window      | <code>exit</code> or <code>Ctrl+b &amp;</code> |

## 11.5 Panes

Panes split a window into sections — each pane is a separate terminal.

| Action                        | Keys                                       |
|-------------------------------|--------------------------------------------|
| Split horizontal (top/bottom) | <code>Ctrl+b "</code>                      |
| Split vertical (left/right)   | <code>Ctrl+b %</code>                      |
| Move between panes            | <code>Ctrl+b arrow</code>                  |
| Close pane                    | <code>exit</code> or <code>Ctrl+b x</code> |
| Toggle full-screen            | <code>Ctrl+b z</code>                      |
| Resize pane                   | <code>Ctrl+b Ctrl+arrow</code>             |

## 11.6 Typical Workflow

1. SSH to server
2. Start tmux: `tmux new -s project`
3. Split into panes for different tasks
4. Work on your task
5. Detach when done: `Ctrl+b d`
6. Disconnect SSH
7. Later: SSH back, `tmux attach -t project`

8. Everything is exactly where you left it

## Example Setup

Start named session, split vertically (editor left, terminal right), split right pane horizontally (terminal top, logs bottom). Result: editor on the left, terminal top right, logs bottom right — all in one SSH connection, surviving disconnection.

```
# Start named session tmux new -s dev # Split vertically Ctrl+b % # Open
editor in left pane vim myfile.py # Move to right pane Ctrl+b right # Split
right pane horizontally Ctrl+b " # Watch logs in bottom pane tail -f
/var/log/syslog
```

## Summary

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- Tmux keeps sessions alive when you disconnect
- Start with `tmux`, detach with **Ctrl+b d**, reattach with `tmux attach`
- Use named sessions: `tmux new -s name`
- Windows are tabs: **Ctrl+b c** creates, **Ctrl+b n/p** navigates
- Panes split windows: **Ctrl+b %** vertical, **Ctrl+b "** horizontal
- The prefix **Ctrl+b** starts all tmux commands

## Key Definitions

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**Tmux:** Terminal multiplexer — manages persistent terminal sessions

**Session:** A tmux instance that persists even when disconnected

**Window:** A tab within a tmux session

**Pane:** A split within a tmux window

**Prefix Key:** `Ctrl+b` — the key combination that starts all tmux commands

**Detach:** Disconnect from tmux while leaving the session running

**Attach:** Reconnect to a running tmux session

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